PHYSICS

Higher Level

Set A | Paper 2

Challenge mock examination

Time allowed: 150 minutes Maximum mark: 90

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A calculator and a clean current Physics data booklet are permitted throughout.

• All structured questions are compulsory. There are no optional topics or optional questions.

• Answer all questions. Show physical reasoning, working and units in written responses.

• Use the constants and assumptions given in a question. Otherwise use the data booklet and retain unrounded intermediate values.

• Give numerical answers to an appropriate number of significant figures. Diagrams are schematic unless explicitly stated to be to scale.

• The number in square brackets gives the maximum mark for a written part.

Thriving Scholars challenge practice | Current Physics syllabus | 2026

2 Question 1 | Question paper

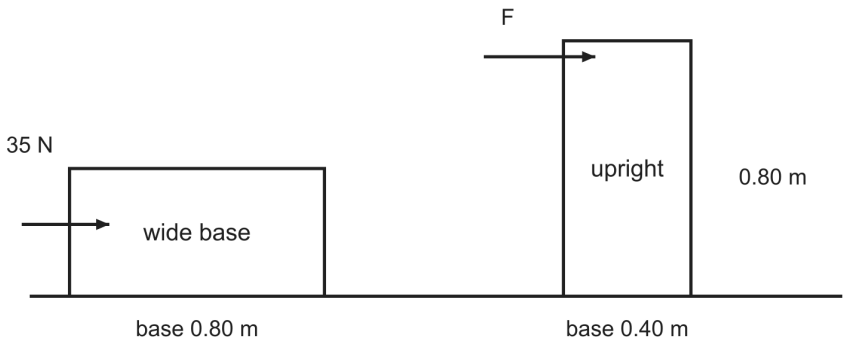
Paper 2 - Structured questions

2 Question 1 [10 marks]

A rigid box of mass $6.00\,\text{kg}$ has a rectangular cross section $0.80\,\text{m}$ by $0.40\,\text{m}$ and its centre of mass at the geometric centre. The horizontal floor has $\mu_s = 0.50$ and $\mu_k = 0.30$. Use $g = 9.81\,\text{m}\,\text{s}^{-2}$.

First the box rests on its $0.80\,\text{m}$ wide base. A horizontal $35.0\,\text{N}$ force acts through its centre and moves it $1.20\,\text{m}$ from rest. The force is removed and the box slides to rest.

In a second experiment the box rests upright on its $0.40\,\text{m}$ wide base. A slowly increasing horizontal force is applied at height $0.80\,\text{m}$.



(a) Calculate the maximum static friction and explain why motion begins in the first experiment. [2]

(b) Determine the speed when the force is removed and the further distance before the box stops. [3]

(c) Determine whether the upright box slides or tips first, and calculate the corresponding applied-force threshold. [3]

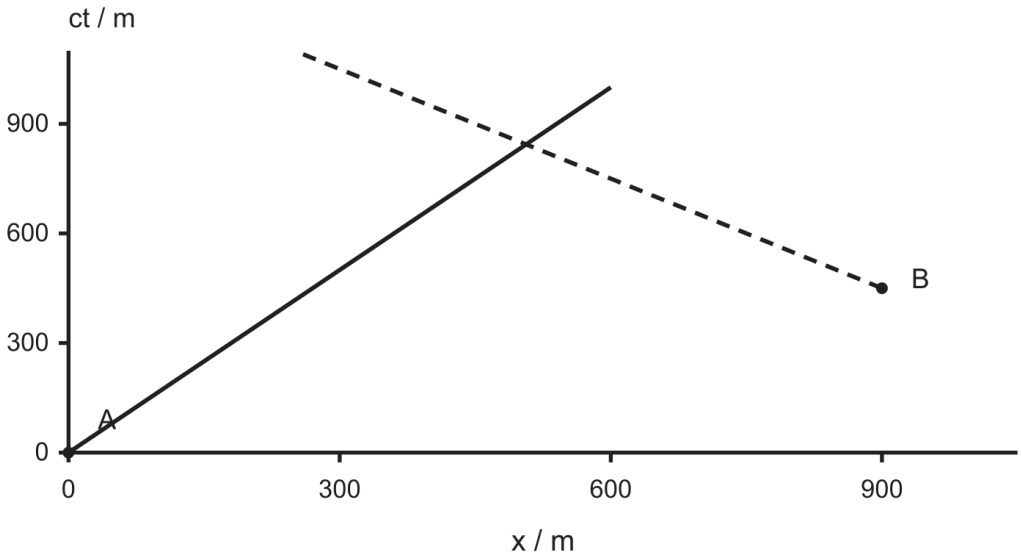
(d) For the upright box, determine the application height at which sliding and tipping thresholds coincide. [2]

2 Question 2 | Question paper

2 Question 2 [10 marks]

Two laboratory beacons lie at $x = 0$ and $x = 900\,\text{m}$. The left beacon emits event A at $t = 0$; the right emits event B at $t = 1.50\,\mu\text{s}$. A relay moves along $+x$ at $0.600c$, passing $x = 0$ at A. Laboratory and relay clocks are zero at A. Use $c = 3.00 \times 10^8\,\text{m}\,\text{s}^{-1}$.

The diagram shows the relay worldline (solid) and the left-going light pulse emitted at B (dashed). In the relay frame use $\Delta x' = \gamma(\Delta x - v\Delta t)$ and $\Delta t' = \gamma(\Delta t - v\Delta x/c^2)$.



(a) Calculate $\Delta x'$ and $\Delta t'$ between emissions A and B. [3]

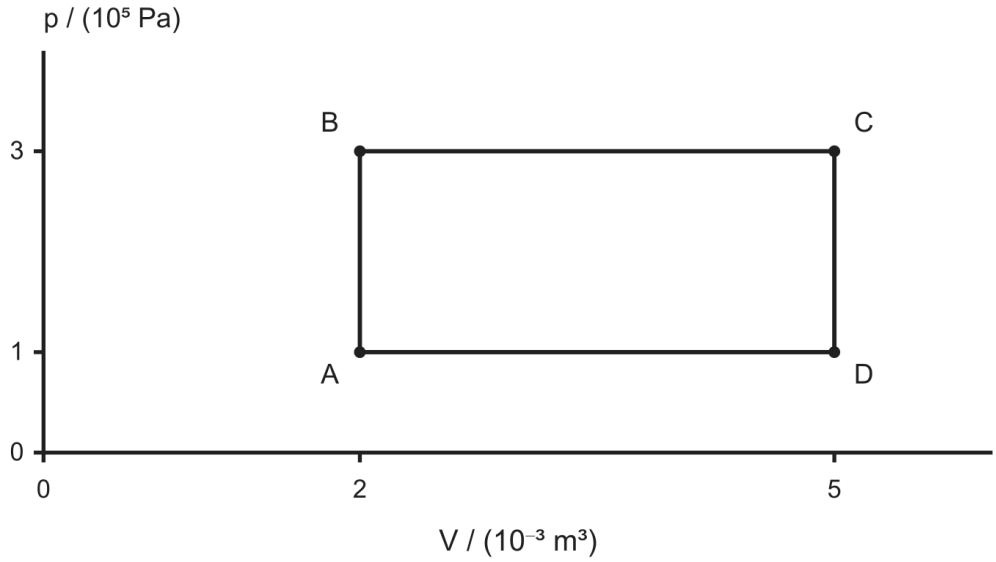
(b) Explain the order of the emissions in the relay frame and whether either emission could cause the other. [2]

(c) Keeping the beacon positions fixed, determine the laboratory time of B that would make the two emissions simultaneous in the relay frame. [2]

(d) For the original emission times, determine the relay-clock reading when it receives the pulse from the right beacon. [3]

2 Question 3 [10 marks]

A closed sample of monatomic ideal gas follows $A \to B \to C \to D \to A$ on the diagram, quasi-statically along each straight segment. It takes in heat from a $1000\,\mathrm{K}$ reservoir during AB and BC, and rejects heat to a $100\,\mathrm{K}$ reservoir during CD and DA. Its amount is $0.200\,\mathrm{mol}$ and $R = 8.31\,\mathrm{J\,mol^{-1}\,K^{-1}}$. Use $U = 3pV/2$ and the convention $Q = \Delta U + W$, with W positive for work by the gas.



(a) Determine the net work by the gas in one cycle and relate its sign to the cycle direction. [2]

(b) Calculate the total heat absorbed from the hot reservoir in one cycle. [3]

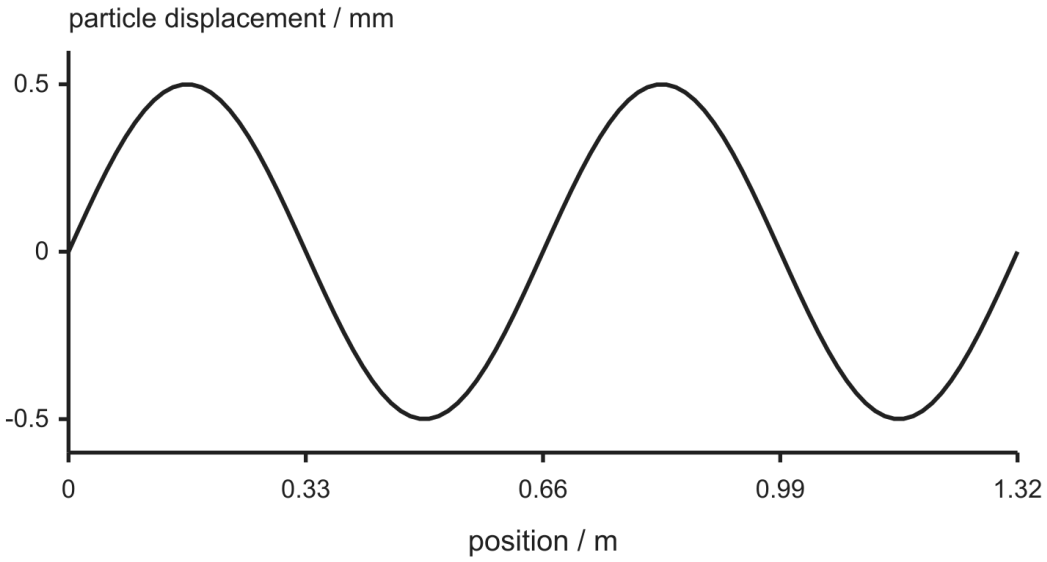
(c) Find the efficiency and compare it with the Carnot limit for the stated reservoirs. [2]

(d) Calculate the total entropy change of the gas plus both reservoirs in one cycle. Explain the physical meaning of its sign. [3]

2 Question 4 [10 marks]

Two stationary sound beacons A and B lie to the left and right of a moving receiver. The air is stationary. The graph is a spatial snapshot of the wave from A, whose period is $2.00\,\text{ms}$. Beacon B emits at $650\,\text{Hz}$. While the receiver moves directly right, its measured frequency ratio is $f_{B'}/f_{A'} = 1.80$. The receiver speed is constant and smaller than the sound speed.

The beacons also transmit an optical line of laboratory wavelength $656.0\,\text{nm}$. The receiver spectrometer resolves wavelength differences no smaller than $0.010\,\text{nm}$. Use $c = 3.00 \times 10^8\,\text{m}\,\text{s}^{-1}$ for light.



(a) Use the graph and period to determine the sound speed and beacon A frequency. [2]

(b) Determine the receiver speed from the measured sound-frequency ratio. [3]

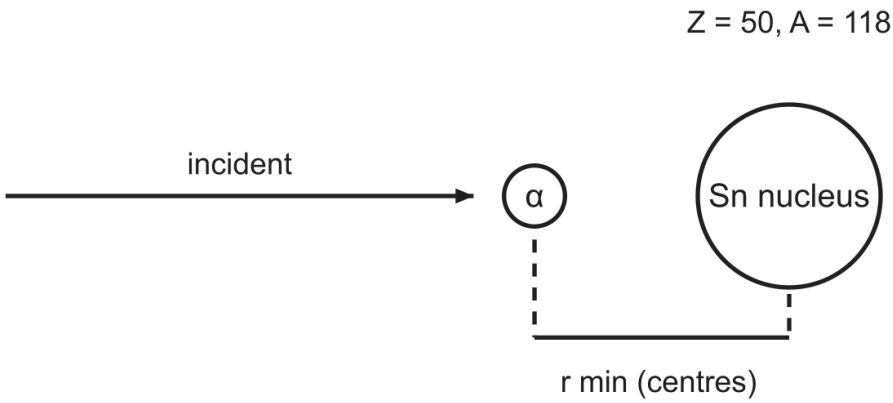
(c) Calculate the two sound frequencies actually received. [2]

(d) Estimate the magnitude of the optical wavelength shift caused by this receiver motion. Decide whether the spectrometer can resolve it and explain why the sound shift is much more noticeable. [3]

2 Question 5 [10 marks]

An initially stationary $^{212}_{84}\mathrm{Po}$ nucleus emits an alpha particle and becomes a lead nucleus. The decay releases $8.95\,\mathrm{MeV}$ of kinetic energy. For recoil calculations take the daughter-to-alpha mass ratio as $208:4$ and use nonrelativistic mechanics.

An alpha particle from this decay is subsequently directed head-on towards a stationary $^{118}_{50}\mathrm{Sn}$ nucleus. For this scattering calculation neglect the tin recoil and use electrostatic repulsion until nuclear-scale distances are reached. Use $k = 8.99 \times 10^{9}\,\mathrm{N\,m^{2}\,C^{-2}}$, $e = 1.60 \times 10^{-19}\,\mathrm{C}$ and $R = 1.20 \times 10^{-15} A^{1/3}\,\mathrm{m}$.



(a) Write the nuclear decay equation and state the momentum relationship between the products. [2]

(b) Determine the kinetic energies of the alpha particle and lead nucleus. [3]

(c) Calculate the alpha particle distance of closest approach to the tin nucleus. [3]

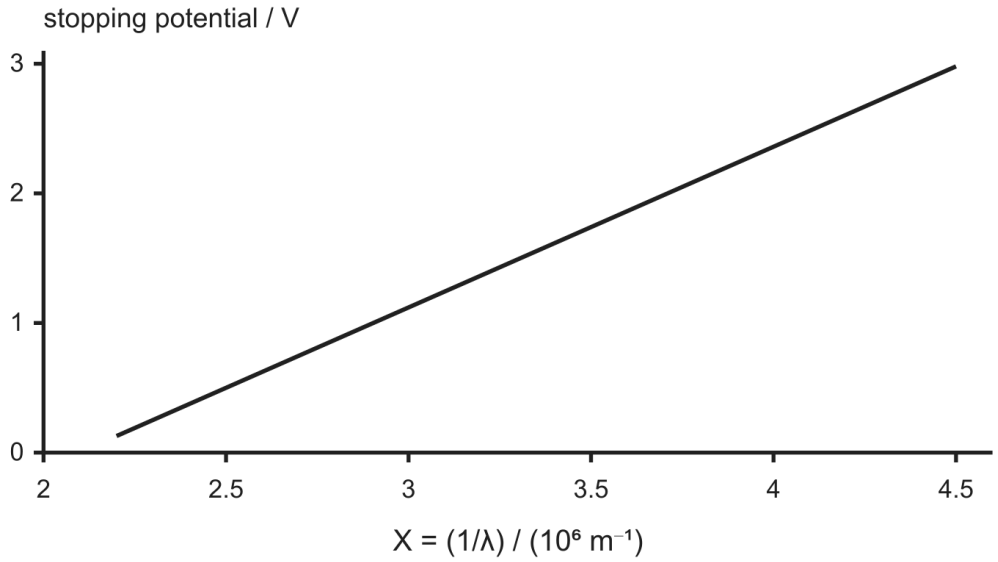
(d) Estimate the incident alpha kinetic energy needed for the electrostatic closest approach to equal the tin nuclear radius. Explain why the simple model should be treated cautiously at that scale. [2]

2 Question 6 [10 marks]

A vacuum photocell is illuminated by monochromatic light. A model experimental fit over its emitting range is

$V_s = (1.240\,\mathrm{V})X - 2.600\,\mathrm{V}, \qquad X = \frac{1/\lambda}{10^6\,\mathrm{m^{-1}}}.$

The plotted V_s is the magnitude of the stopping potential. Use $e = 1.60 \times 10^{-19}\,\mathrm{C}$ and $c = 3.00 \times 10^8\,\mathrm{m\,s^{-1}}$. In part(d) use accepted values $h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$ and $m_e = 9.11 \times 10^{-31}\,\mathrm{kg}$. Assume independent single-photon emission and no space-charge limitation.



(a) Use the fit and the photoelectric equation to determine the experimental Planck constant and the metal work function in electronvolts. [3]

(b) Determine the threshold wavelength predicted by the fit. [2]

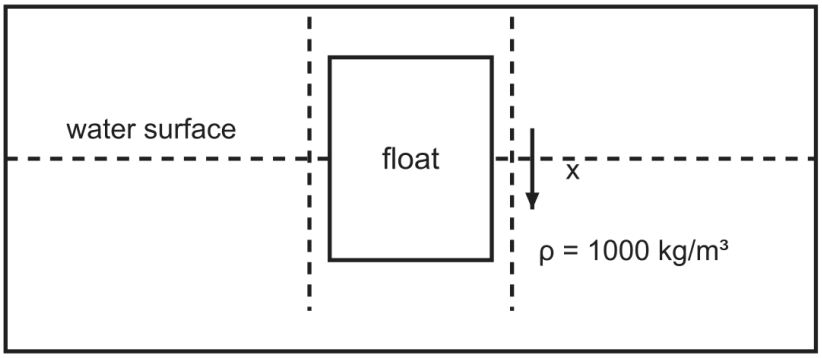
(c) At $300\,\mathrm{nm}$ the optical power is increased by a factor4 without changing wavelength. Predict the saturation-current and stopping-potential changes, with reasons. [2]

(d) Determine the de Broglie wavelength of the most energetic electrons emitted under $300\,\mathrm{nm}$ illumination. [3]

2 Question 7 [10 marks]

A uniform vertical cylinder has mass $0.900\,\text{kg}$, height $0.300\,\text{m}$ and constant cross-sectional area $6.00 \times 10^{-3}\,\text{m}^2$. A frictionless vertical guide prevents tipping without exerting a vertical force. It floats in water of density $1000\,\text{kg}\,\text{m}^{-3}$. Use $g = 9.81\,\text{m}\,\text{s}^{-2}$. For this model ignore damping, surface tension and the inertia of the surrounding water. Displacements remain small enough that the cylinder is neither fully submerged nor lifted clear of the water. Take downward displacement from equilibrium as x.

The cylinder is released from rest at amplitude $0.0400\,\text{m}$. A later run adds a $0.300\,\text{kg}$ payload fixed rigidly to its top and repeats the same displacement from the new equilibrium.



Side view; displacement x positive downward

(a) Determine the equilibrium submerged length before the payload is added. [2]

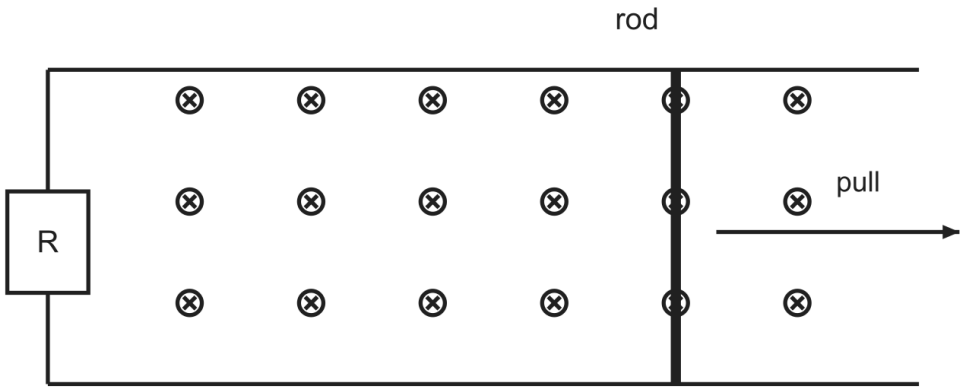
(b) Show that the restoring acceleration is proportional to $-x$ and calculate the original period. [3]

(c) Calculate the original maximum speed and total oscillation energy. [2]

(d) Predict the loaded period. Check explicitly whether the stated amplitude still keeps the cylinder partly submerged throughout the loaded run. [3]

2 Question 8 [10 marks]

A conducting rod of length $L = 0.500\,\text{m}$ and mass $0.200\,\text{kg}$ slides right on horizontal conducting rails. The circuit resistance is $0.800\,\Omega$, concentrated in the fixed resistor; rail and rod resistance are negligible. A uniform constant magnetic field $0.400\,\text{T}$ is directed into the page. A constant horizontal pulling force of $0.100\,\text{N}$ acts right. Ignore friction and self-induction.



B into page; rail separation L

(a) At a rod speed of $1.00\,\text{m}\,\text{s}^{-1}$, calculate the induced emf and current and state the current direction through the rod. [3]

(b) Determine the rod acceleration at this speed. [2]

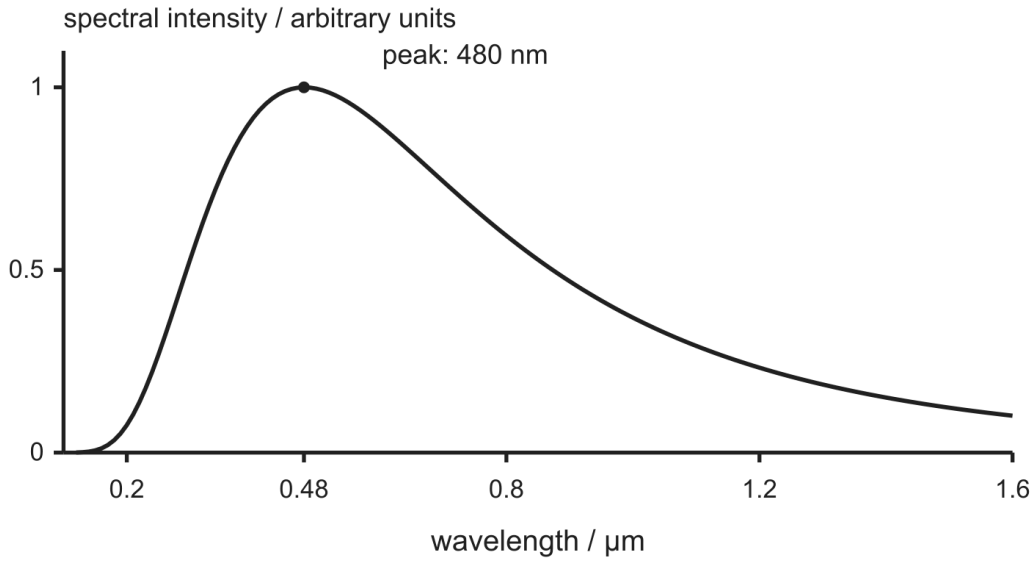
(c) Determine the terminal speed and the steady electrical power dissipated. Explain the corresponding mechanical energy balance. [3]

(d) In a separate run with a constant field of $0.800\,\text{T}$, determine the acceleration when speed is $2.00\,\text{m}\,\text{s}^{-1}$ and the new terminal speed. Other parameters are unchanged. [2]

2 Question 9 [10 marks]

A star at distance $12.0\,\text{pc}$ has apparent brightness $1.20 \times 10^{-8}\,\text{W m}^{-2}$. Its continuum spectrum has the maximum shown. Model it as a spherical black body. Use $1\,\text{pc} = 3.086 \times 10^{16}\,\text{m}$, Wien constant $2.90 \times 10^{-3}\,\text{m K}$, $\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$ and $c = 3.00 \times 10^{8}\,\text{m s}^{-1}$.

Suppose its luminosity is sustained by hydrogen fusion in which 0.700% of the processed hydrogen rest mass is converted to released energy. For comparison, the Sun has radius $6.96 \times 10^{8}\,\text{m}$ and effective temperature approximately $5800\,\text{K}$.



(a) Determine the effective temperature and luminosity of the star. [2]

(b) Calculate its radius. [2]

(c) Determine the rate at which hydrogen mass must be processed, distinguishing it from the rate of rest-mass conversion to energy. [3]

(d) Compare the star with the Sun on an HR diagram and explain the physical balance that permits a stable stellar radius. State one limit of inferring an exact evolutionary stage from the supplied observations. [3]

PHYSICS

Higher Level

Set A | Paper 2

Worked solution book

Complete the examination before consulting this book. Marking points indicate a method (M), accuracy (A) or reasoning (R) award. The number after the code gives its value. Award only points supported by the candidate's retained working.

Accept valid equivalent approaches and reasonable rounding. Apply error carried forward only where the subsequent method remains physically meaningful; do not reward a contradiction of stated conditions. A correct multiple-choice letter earns one mark; no negative marks apply.

MCQs are automatically marked on the portal. Written marks are self-assessed. Reports show these separately and do not calculate an official IB grade.

Paper 2 - Worked solutions

2 Question 1 [10 marks]

(a) [2 marks]

A1 $F_{s,max} = \mu_s mg = 29.43\,\text{N}$.

R1 The applied $35.0\,\text{N}$ exceeds this threshold, so static friction cannot maintain equilibrium; the centre-height force does not overturn the wide box first.

(b) [3 marks]

M1 $F_k = 17.658\,\text{N}$ and work-energy gives $\frac{1}{2}mv^2 = (35.0 - 17.658)(1.20)$.

A1 $v = 2.63378\,\text{m s}^{-1} \approx 2.63\,\text{m s}^{-1}$.

A1 After release, $F_k d = \frac{1}{2}mv^2$, giving $d = 1.17853\,\text{m} \approx 1.18\,\text{m}$.

(c) [3 marks]

M1 At incipient tipping about the downstream lower corner, $F(0.80) = mg(0.20)$.

A1 $F_{tip} = 14.715\,\text{N} \approx 14.7\,\text{N}$.

R1 This is less than the sliding threshold $29.43\,\text{N}$, so the upright box tips first. Static friction at this force is below its limiting value.

(d) [2 marks]

M1 Set $mg(0.20)/H = \mu_s mg$.

A1 $H = 0.20/0.50 = 0.400\,\text{m}$. A lower push produces sliding first; a higher push produces tipping first.

2 Question 2 [10 marks]

(a) [3 marks]

A1 $\gamma = 1/\sqrt{1 - 0.600^2} = 1.25.$

A1 $\Delta x\prime = 1.25(900 - 0.600 \times 450) = 787.5\,\text{m}.$

A1 $\Delta t\prime = 1.25(1.50 - 1.80) = -0.375\,\mu\text{s}.$

(b) [2 marks]

R1 Because $\Delta t\prime < 0$, the relay assigns B before A.

R1 The interval is spacelike: $(c\Delta t)^2 - (\Delta x)^2 = 450^2 - 900^2 < 0$. No light or slower signal can connect the two emissions, so reversing their coordinate order violates no causality.

(c) [2 marks]

M1 Simultaneity requires $\Delta t = v\Delta x/c^2$.

A1 $t_B = 0.600(900)/c = 1.80\,\mu\text{s}$ after A in the laboratory.

(d) [3 marks]

M1 The relay has $x = 0.600ct$ and the incoming pulse has $x = 900 - c(t - 1.50 \times 10^{-6})$.

M1 Equating gives $t = 2.8125\,\mu\text{s}$ and $x = 506.25\,\text{m}$ in the laboratory.

A1 The relay proper time from A is $t/\gamma = 2.25\,\mu\text{s}$.

2 Question 3 [10 marks]

(a) [2 marks]

M1 The clockwise cycle encloses $\Delta p \Delta V = (2.00 \times 10^5)(3.00 \times 10^{-3})$.

A1 $W_{cycle} = +600\,\text{J}$; expansion at high pressure does more work than compression at low pressure.

(b) [3 marks]

M1 Along AB, $W = 0$ and $Q_{AB} = \frac{3}{2} V_A (p_B - p_A) = 600\,\text{J}$.

M1 Along BC, $Q = \Delta U + p\Delta V = (\frac{3}{2} + 1) p_B (V_C - V_B) = 2250\,\text{J}$.

A1 $Q_{in} = 2850\,\text{J}$.

(c) [2 marks]

A1 $\eta = W/Q_{in} = 600/2850 = 0.210526 \approx 21.1\%$.

R1 The Carnot bound is $1 - 100/1000 = 0.900$, so the calculated efficiency is below it.

(d) [3 marks]

M1 The gas returns to its initial state, so $\Delta S_{gas} = 0$. Rejected heat is $Q_{out} = 2850 - 600 = 2250\,\text{J}$.

M1 $\Delta S_{total} = -2850/1000 + 2250/100 = 19.65\,\text{J K}^{-1}$.

R1 The positive total entropy change is consistent with an irreversible cycle, including heat transfer across finite temperature differences. Quasi-static gas motion alone does not make the whole process reversible.

2 Question 4 [10 marks]

(a) [2 marks]

A1 The wavelength is $0.660\,\text{m}$, so $v = \lambda/T = 0.660/0.00200 = 330\,\text{m s}^{-1}$.

A1 $f_A = 1/T = 500\,\text{Hz}$.

(b) [3 marks]

M1 The receiver recedes from A and approaches B, so $f_A\prime = 500(v-u)/v$ and $f_B\prime = 650(v+u)/v$.

M1 $650(330+u)/[500(330-u)] = 1.80$.

A1 $u = 53.2258\,\text{m s}^{-1} \approx 53.2\,\text{m s}^{-1}$ to the right.

(c) [2 marks]

A1 $f_A\prime = 419.355\,\text{Hz} \approx 419\,\text{Hz}$.

A1 $f_B\prime = 754.839\,\text{Hz} \approx 755\,\text{Hz}$, retaining the ratio 1.80.

(d) [3 marks]

M1 $|\Delta\lambda|/\lambda \approx u/c = 1.77419 \times 10^{-7}$.

A1 $|\Delta\lambda| = 0.0001164\,\text{nm}$, far below $0.010\,\text{nm}$ and therefore unresolved.

R1 The dimensionless motion is $u/v \approx 0.161$ for sound but only $u/c \approx 1.77 \times 10^{-7}$ for light; the same speed gives very different fractional shifts.

2 Question 5 [10 marks]

(a) [2 marks]

A1 $^{212}_{84}\mathrm{Po} \rightarrow {}^{208}_{82}\mathrm{Pb} + {}^{4}_{2}\mathrm{He}$.

R1 The momenta are equal in magnitude and opposite in direction because the parent initially has zero momentum.

(b) [3 marks]

M1 With equal p, $K = p^2/(2m)$ gives $K_\alpha/K_{Pb} = 208/4 = 52$.

M1 The energies sum to $8.95\,\mathrm{MeV}$, so $K_\alpha = 8.95(208/212)$ and $K_{Pb} = 8.95(4/212)$.

A1 $K_\alpha = 8.78113\,\mathrm{MeV} \approx 8.78\,\mathrm{MeV}$ and $K_{Pb} = 0.168868\,\mathrm{MeV} \approx 0.169\,\mathrm{MeV}$.

(c) [3 marks]

M1 At closest approach $K_\alpha = k(2e)(50e)/r_{min}$.

M1 Convert $8.78113\,\mathrm{MeV}$ to joules using $10^6 e$.

A1 $r_{min} = 1.63806 \times 10^{-14}\,\mathrm{m} \approx 1.64 \times 10^{-14}\,\mathrm{m}$.

(d) [2 marks]

M1 $R_{Sn} = 1.20 \times 10^{-15}(118)^{1/3} = 5.88584 \times 10^{-15}\,\mathrm{m}$ and $K = k(100e^2)/R_{Sn} \approx 24.4\,\mathrm{MeV}$.

R1 This extrapolation reaches the finite-sized nucleus, where the point-charge/electrostatic-only approximation can fail and nuclear interactions become relevant; it is a scale estimate, not a guaranteed scattering prediction.

2 Question 6 [10 marks]

(a) [3 marks]

M1 From $eV_s = hc/\lambda - \Phi$ and $1/\lambda = 10^6 X$, the fitted slope is $10^6 hc/e$.

A1 $h_{exp} = 1.240e/(10^6 c) = 6.61333 \times 10^{-34}\,\text{J s}$.

A1 The intercept gives $\Phi/e = 2.600\,\text{V}$, hence work function $2.600\,\text{eV}$.

(b) [2 marks]

M1 Set $V_s = 0$: $X_0 = 2.600/1.240 = 2.09677$.

A1 $\lambda_0 = 1/(10^6 X_0) = 4.76923 \times 10^{-7}\,\text{m} \approx 477\,\text{nm}$.

(c) [2 marks]

R1 At fixed wavelength, four times the power means four times the photon arrival rate and therefore four times the saturation current under the assumptions.

R1 The photon energy and work function are unchanged, so the maximum electron kinetic energy and stopping potential are unchanged.

(d) [3 marks]

M1 The fit at $X = 1/(300 \times 10^{-9} \times 10^6) = 3.33333$ gives $V_s = 1.53333\,\text{V}$ and $K_{max} = eV_s$.

M1 $p = \sqrt{2 m_e K_{max}}$ and $\lambda_{dB} = h/p$.

A1 $\lambda_{dB} \approx 9.92 \times 10^{-10}\,\text{m} = 0.992\,\text{nm}$, using unrounded values.

2 Question 7 [10 marks]

(a) [2 marks]

M1 Equilibrium requires $\rho g A D = mg$.

A1 $D = m/(\rho A) = 0.150\,\text{m}$.

(b) [3 marks]

M1 A downward displacement adds buoyancy $\rho g A x$, so $F_{net} = -\rho g A x$ and $a = -(\rho g A/m)x$.

M1 $\omega = \sqrt{\rho g A/m} = \sqrt{58.86/0.900}$ and $T = 2\pi/\omega$.

A1 $T \approx 0.777\,\text{s}$.

(c) [2 marks]

A1 $v_{max} = \omega A_{osc} \approx 0.323\,\text{m s}^{-1}$.

A1 $E = \frac{1}{2}\rho g A A_{osc}^2 = 0.047088\,\text{J} \approx 0.0471\,\text{J}$.

(d) [3 marks]

M1 The total oscillating mass is $1.200\,\text{kg}$ while $\rho g A$ is unchanged, so $T_{loaded} = 2\pi\sqrt{1.200/58.86}$.

A1 $T_{loaded} \approx 0.897\,\text{s}$.

R1 The new draft is $1.200/(1000 \times 0.00600) = 0.200\,\text{m}$. During oscillation it ranges from 0.160 to $0.240\,\text{m}$, strictly between 0 and the $0.300\,\text{m}$ height, so the stated model remains geometrically valid.

2 Question 8 [10 marks]

(a) [3 marks]

A1 $\mathcal{E} = BLv = 0.200\,\text{V}$.

A1 $I = \mathcal{E}/R = 0.250\,\text{A}$.

R1 Positive charges are driven upward along the moving rod; the induced current is upward through the rod, giving a leftward magnetic force.

(b) [2 marks]

M1 $F_B = BIL = 0.0500\,\text{N}$ left, opposing the rightward pull.

A1 $a = (0.100 - 0.0500)/0.200 = 0.250\,\text{m s}^{-2}$ right.

(c) [3 marks]

M1 Terminal balance gives $F = B^2L^2v_t/R$, hence $v_t = FR/(B^2L^2) = 2.00\,\text{m s}^{-1}$.

A1 $P = Fv_t = 0.200\,\text{W}$, also I^2R with $I = 0.500\,\text{A}$.

R1 At terminal speed kinetic energy is constant; all pulling power is converted to resistor heating through electromagnetic induction.

(d) [2 marks]

A1 $a = [0.100 - (0.800)^2(0.500)^2(2.00)/0.800]/0.200 = -1.50\,\text{m s}^{-2}$, opposite the motion.

A1 The terminal speed is $0.500\,\text{m s}^{-1}$, one quarter of the original because magnetic drag is proportional to B^2v.

2 Question 9 [10 marks]

(a) [2 marks]

A1 $T = 2.90 \times 10^{-3} / (480 \times 10^{-9}) = 6041.67\,\text{K} \approx 6.04 \times 10^{3}\,\text{K}$.

A1 $L = 4\pi d^2 b \approx 2.07 \times 10^{28}\,\text{W}$, converting $12.0\,\text{pc}$ to metres before squaring.

(b) [2 marks]

M1 $R = \sqrt{L/(4\pi\sigma T^4)}$.

A1 $R \approx 4.67 \times 10^{9}\,\text{m}$, about 6.7 solar radii.

(c) [3 marks]

M1 The rate of mass actually converted to energy is $L/c^2 \approx 2.30 \times 10^{11}\,\text{kg s}^{-1}$.

M1 Only a fraction 0.00700 of the processed hydrogen is converted, so $\dot{m}_H = L/(0.00700c^2)$.

A1 $\dot{m}_H \approx 3.28 \times 10^{13}\,\text{kg s}^{-1}$, rather than L/c^2.

(d) [3 marks]

R1 The star is slightly hotter and far more luminous than the Sun, placing it above and slightly left of the Sun on a conventional HR diagram; its much larger radius is consistent with an evolved enlarged star rather than a Sun-sized object.

R1 In a stable star, inward gravitational pressure is balanced by outward thermal pressure and/or radiation pressure, with fusion providing the energy supply.

R1 A precise evolutionary stage is not fixed by temperature and luminosity alone: stellar mass, composition, age or additional spectroscopic/evolutionary evidence is needed. Accept a physically justified equivalent limitation.

End of worked solution book